Partial Kinematic Synthesis of a Finger Exoskeleton Mechanism Using Differential Evolution and Genetic Algorithm

***Abstract*—This work uses two optimization techniques, Differential Evolution (DE) and the Genetic Algorithm (GA), to obtain the *partial* dimensional synthesis of a finger exoskeleton mechanism, fitting the movement up to the medial phalanx (distal interphalangeal joint, DIP). The problem is formulated with 16 design parameters and an objective function based on the weighted Chamfer distance with respect to 120 motion-capture (MOCAP) points of a fine pinch grasp, taken from the public hand kinematics database [1], to which a penalty for movement reversals and a term that favors small dimensions are added. The hyperparameters of both algorithms are tuned symmetrically with the Optuna tool, using the same seed and the same number of 25 trials per method. Under this setup, with the same objective function, the same bounds and the same number of generations, DE reaches a global error of 2.17 mm versus 2.55 mm for the GA and produces a more compact mechanism (sum of structural links of 347.2 mm against 457.1 mm). The GA, however, obtains a lower error at the DIP joint (1.43 mm versus 1.53 mm), while DE keeps the advantage in the global error and in the proximal phalanx (PIP). The results indicate that both techniques are suitable for this class of dimensional synthesis and that the specific choice must optimize the trade-off between fitting error and mechanism compactness.**

***Index Terms*—differential evolution, genetic algorithm, dimensional synthesis, hand exoskeleton, dimensional optimization, mechanism chain.**

I. Introduction

HAND exoskeletons are robotic assistance and rehabilitation devices whose function is to accompany or help recover the movement of the fingers [2]. For the assistance to be comfortable and safe, the mechanism must faithfully reproduce the natural joint trajectories of the finger over the whole flexion range; otherwise, unwanted reaction forces appear at the joints and the device stops being comfortable for the user [3]. Achieving that tracking depends largely on correctly choosing the link lengths of the mechanism, a problem known as *dimensional synthesis* [4].

Dimensional synthesis for path generation is a hard optimization problem: the solution space contains large regions where the mechanism simply does not assemble or reverses its motion, and the relation between the link lengths and the curve described by the mechanism is nonlinear and has many local optima. For this reason, classical gradient-based methods tend to get trapped in suboptimal solutions, which motivates the use of population-based metaheuristics such as Differential Evolution (DE) and Genetic Algorithms (GA), which can traverse the search space without needing derivative information [4], [5].

Although both optimization techniques are widely used, their effectiveness depends on the nature of the problem, on how the variables are represented and on the tuning of the hyperparameters [6], [7]. The choice between one or the other is rarely obvious and almost always requires testing them on the specific case to be solved.

Dimensional synthesis of linkages for path generation has historically been addressed with analytical methods and, more recently, with metaheuristics that avoid the limitations of gradient-based approaches on nonconvex and discontinuous functions [4], [5]. Four-bar mechanisms are a recurring building block both in path synthesis and in finger exoskeletons, due to their simplicity and their ability to approximate complex coupler curves with few parameters [3], [8]. When several four- and five-bar stages are chained, the number of variables grows and the search space becomes strongly multimodal, which makes the use of global optimization techniques unavoidable.

In the field of hand exoskeletons, several recent works use swarm-based or evolutionary optimization to tune the dimensions of mechanisms that reproduce the natural flexion of the fingers [2], [8]. The choice of algorithm, however, is rarely justified empirically on the specific mechanism. The direct comparison between families of algorithms has also been studied: in continuous-domain problems, DE and particle swarm optimization tend to behave more naturally than the GA, which was originally conceived for discrete representations [6], [7]. Nevertheless, the outcome of the comparison depends on the encoding, on the operators and on the tuning of hyperparameters, so it remains necessary to characterize the behavior on real instances [9], [10].

Unlike those works, here the same formulation, objective function and number of generations are fixed for both algorithms on a real exoskeleton mechanism, and not only the fitting error is evaluated but also the coherence and compactness of the resulting mechanism, two aspects that are decisive for its later manufacture.

In this work both algorithms are used under equal conditions, that is, starting from the same formulation, the same objective function and the same number of generations, so that any difference in the result is due to the algorithm and not to how the experiment was set up. This with the aim of determining the best technique to synthesize planar mechanisms.

For the comparison to be reproducible, a *partial* version of the mechanism is optimized, which generates the movement up to the medial phalanx (PIP joint). This case is smaller than the full three-phalanx model, but it keeps the essential difficulties of the problem (the coupling between stages, the assembly constraints and the need for the movement not to reverse), which makes it possible to study the effect of the optimization algorithm without the added complexity of the full model.

A. Contributions

The contributions of this article are: (i) a formulation of the partial dimensional synthesis problem with 16 parameters, an objective function based on the weighted Chamfer distance and feasibility penalties; (ii) a controlled comparison of DE and GA under identical conditions, which evaluates not only the fitting error but also the coherence and compactness of the resulting mechanism; and (iii) an analysis of the reasons why DE keeps a slight advantage in this continuous-variable problem.

II. Description of the Exoskeleton Mechanisms

To the original model [] a third four-bar stage is added, in order to correct tracking problems that it presents at the DIP joint. Since in this work the partial optimization of the chain is carried out, the third four-bar mechanism is omitted. The present study focuses on the dimensional optimization of the resulting mechanism (Figure 1).

The device reproduces the flexion of the finger through a chain of coupled stages driven by a single input crank, which makes it possible to govern all the joints with a single active degree of freedom. The first stage combines a five-bar mechanism (links L_1, L_2, L_3, L_4 and the ground B_1) and a four-bar one (links L_4, L_5, h_{sp}, d_{sp} and the ground B_2; the supports h_{sp} and d_{sp} form a right triangle whose hypotenuse acts as a single equivalent link) that transfer the movement from the metacarpophalangeal (MCP) joint to the proximal phalanx (proximal interphalangeal joint, PIP). The second stage repeats the five-bar pattern (links L_5, L_3, L_6, L_7 and the ground with length $F_p - 2\,d_{sp}$) plus four-bar (links L_7, L_8, c_2 and the distance from the PIP to h_{sp}) to carry the movement from the PIP to the medial phalanx (DIP joint). The proximal phalanx F_p and the medial phalanx F_m form the finger line (MCP, PIP and DIP joints), while the chain of links L_1–L_8 is placed on the back of the hand. The grounds B_1 and B_2 are indicated, as well as the rocker c_2, the displacements of the support h_{sp} and d_{sp}, and the auxiliary angle $\theta_{aux_{fm}}$ that orients the medial phalanx with respect to the rocker c_2. The revolute joints are drawn as white circles and the fixed supports with hatching. The model stops at the medial phalanx; the third four-bar mechanism and the distal phalanx are not part of this version. The *full* model additionally incorporates a third four-bar stage that locates the distal phalanx; in this study the simplified model is used, which analyzes up to the medial phalanx. This choice reduces the instance without losing the essential difficulties of the problem (stage coupling, assembly constraints and movement monotonicity).

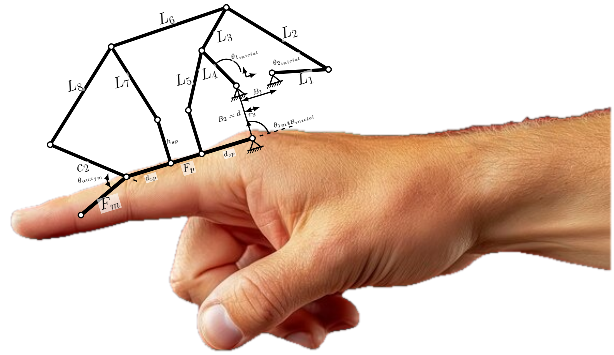

Figure 1. Simplified exoskeleton mechanism mounted on a symbolic finger: the chain of links L_1–L_8 is placed on the back of the hand and the phalanges F_p and F_m are aligned with the proximal and medial phalanges of the finger. The shape of the links corresponds to the mechanism before optimizing.

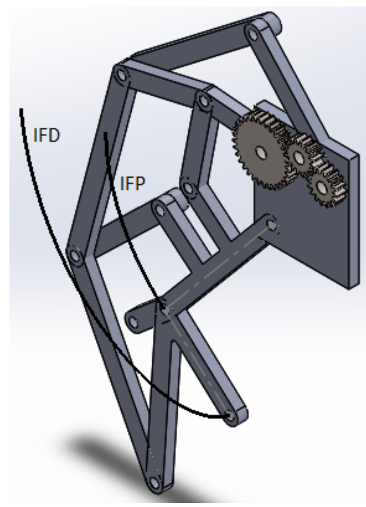

Figure 2. 3D model of the mechanism before optimizing, at the end of its travel, with the trajectories followed by the PIP and DIP joints.

The lengths of the phalanges are considered fixed and are taken from the measurements of a real human finger: proximal phalanx $L_{FP} = 49$ mm, medial phalanx $L_{FM} = 26$ mm and distal phalanx $L_{FD} = 24$ mm (the latter only in the full model). Given a crank position θ_2, the input of the first stage is obtained through a gear ratio r_g and an offset θ_{off} as $\theta_1 = \theta_2/r_g + \theta_{\text{off}}$. The direct kinematics sequentially solves the four-bar loops and propagates the positions of the PIP and the DIP along the whole crank sweep. When a configuration does not assemble, produces non-finite values or generates a reversal of the medial phalanx (non-monotonicity of the DIP angle), the pose is marked as infeasible and discarded. Figure 2 shows the 3D model of the mechanism before optimizing at the end of its travel, together with the trajectories described by the PIP and DIP joints during the full crank sweep. Figure 1 shows the simplified exoskeleton mechanism mounted on a symbolic finger. The dimensional synthesis described in the following sections adjusts these lengths up to the reported solutions.

III. Formulation of the optimization problem

A. Design variables

The design vector $\mathbf{p} \in \mathbb{R}^{16}$ groups the two grounds, the eight links L_1–L_8, the rocker c_2 of the second four-bar stage, the displacements of the support h_{sp} and d_{sp}, the auxiliary angle of the medial phalanx θ_{aux}^{FM}, the gear ratio r_g and the input angular offset θ_{off}. Table I summarizes the search bounds. The length of each link is bounded to 60 mm at most to favor a compact and manufacturable mechanism, and the auxiliary angle θ_{aux}^{FM} is limited to 70° to avoid an anatomically infeasible hyperflexion of the medial phalanx that would cause mechanical interference with the user's finger.

Table I
Design variables and search bounds

Parameter	Min.	Max.
B_1, B_2 (mm)	15	60
L_1–L_8 (mm)	10	60
c_2 (mm)	10	55
h_{sp} (mm)	5	30
d_{sp} (mm)	5	23
θ_{aux}^{FM} (deg)	0	70
r_g (gear ratio)	1.0	8.0
θ_{off} (rad)	$-\pi$	π

B. Objective function

The fit is measured with the symmetric Chamfer distance [11] between the motion-capture point cloud M and the simulated trajectory S. The set of reference points was obtained from the public database [1]. Before evaluating the error, the simulated trajectory is rigidly aligned (rotation R and translation $\mathbf{t}$) with the data through a Procrustes solution [12], so that the error reflects the *shape* of the curve and not its absolute placement in the plane:

$$d_C(M, S) = \frac{1}{|M|}\sum_{\mathbf{m}\in M}\min_{\mathbf{s}\in S}\|\mathbf{m}-\mathbf{s}\| + \frac{1}{|S|}\sum_{\mathbf{s}\in S}\min_{\mathbf{m}\in M}\|\mathbf{s}-\mathbf{m}\|. \tag{1}$$

where M is the set of motion-capture (MOCAP) points, S is the set of points of the simulated trajectory, $\mathbf{m}$ and $\mathbf{s}$ are individual points of each set. The first term penalizes MOCAP points without a nearby simulated point and the second penalizes simulated points that move away from the target curve; their sum is a metric that is robust to sampling differences between the two curves.

The objective function combines the shape error of the two trajectories (that of the PIP and that of the DIP), a monotonicity penalty P_{mono} on the DIP to avoid movement reversals, and two soft regularization terms that push toward small dimensions and auxiliary angle without degrading the fit. The term P_{mono} penalizes trajectory inversions (hooks or loops) in which the exoskeleton mechanism moves backward over its path; its inclusion guarantees that the simulated trajectory progresses monotonically without forming loops that would cause jamming in the mechanism or unwanted forces on the finger:

$$f(\mathbf{p}) = w_{IFP}\, d_C(M_{IFP}, S_{IFP}) + w_{IFD}\, d_C(M_{IFD}, S_{IFD}) + w_m\, P_{\text{mono}}(S_{IFD}) + w_d \sum_i L_i + w_a \frac{\theta_{aux}^{FM}}{\pi}. \tag{2}$$

where $d_C(M_{IFP}, S_{IFP})$ and $d_C(M_{IFD}, S_{IFD})$ are the Chamfer distances between the MOCAP points and the simulated trajectory at the PIP and DIP joints respectively, $P_{\text{mono}}(S_{IFD})$ is the penalty for trajectory inversions at the DIP, $\sum_i L_i$ is the sum of the lengths of the structural links, and θ_{aux}^{FM} is the auxiliary angle of the medial phalanx normalized by π. The weights are $w_{IFP} = 0.40$, $w_{IFD} = 0.60$, $w_m = 5.0$ and $w_d = w_a = 5 \times 10^{-4}$. A greater weight is assigned to the DIP because it is the joint farthest from the input and, therefore, the hardest to fit: it accumulates the errors of the two stages. The weight values are adjusted through iterative experimental tests until an adequate balance between the shape error, the reversal penalty and the regularization is obtained. Infeasible configurations (without assembly or with reversal) receive a fixed penalty of 1000, which creates wide plateaus in the search space and reinforces the multimodal character of the problem. The formulation, the bounds and the objective function are identical for DE and GA, so that the comparison measures the algorithm and not the formulation.

Table II summarizes the kinematic constraints applied during the evaluation of each individual. If any of these conditions is violated, the configuration is considered infeasible and receives the fixed penalty of 1000.

Table II
Kinematic constraints applied in both algorithms

Constraint	Penalty condition
No assembly	The mechanism does not close in some pose
Non-finite values	NaN or infinity is generated
Medial phalanx reversal	Non-monotonicity of $\theta_{fm} > 0.5°$
Minimum lengths	Some link ≤ 5 mm
Support geometry	$F_p - 2\,d_{sp} \leq 1$ mm
Out-of-range positions	Some coordinate > 300 mm

IV. Evaluation of the optimization techniques

Both algorithms are run with the same random seed (42), a conventional value widely adopted in previous evolutionary optimization studies that guarantees the reproducibility of the results, and the same maximum number of generations (300), a budget validated as sufficient given that DE records its last improvement near generation 257 and the GA near generation 268, so that both exploit the full budget without premature stagnation. To avoid the performance depending on hyperparameters chosen by hand – in the case of DE: mutation strategy, population size, mutation factor range and recombination rate; in the case of the GA: population size, crossover and mutation probabilities, SBX and polynomial distribution indices, and tournament size – a common source of bias when comparing metaheuristics, the hyperparameters of each algorithm are tuned automatically and symmetrically with Optuna [13], using the tree-structured Parzen estimator sampler (TPE). The procedure is identical for DE and the GA: in a first stage, Optuna performs 25 trials per algorithm, the same number for both, with short runs that evaluate the fitness reached for different combinations of hyperparameters; in a second stage, the best combination found is used in a deep run of 300 generations, whose results are the ones reported. The Optuna search uses the same seed (42) in both cases, so that the tuning is reproducible and fair. Table III summarizes the resulting hyperparameters. The reported results were obtained with Python and the libraries NumPy 2.0.2, SciPy 1.13.1 and Optuna 4.9.0; it is worth warning that the figures may differ slightly when using other versions of these libraries, since Optuna's TPE sampler may select different hyperparameters for the same seed and SciPy may alter the trajectory of the differential evolution, so that the exact reproduction requires fixing those versions.

A. Differential Evolution

Differential Evolution is a population-based technique for optimization in continuous spaces, whose variants and applications have been the subject of extensive reviews [14], [15]. In this work the `differential_evolution` implementation of SciPy with the `best1bin` strategy is used, which is an open-source library for Python that implements the differential evolution algorithm for the global optimization of nonlinear and multimodal functions. The hyperparameters tuned by Optuna are a relative population size popsize $= 28$, a differential mutation factor that varies randomly in each generation within the range $[0.54, 0.85]$ and recombination 0.77, with tolerance 10^{-7}. In each generation, the trial vectors are built from the best individual and from the scaled difference between two randomly chosen individuals. This differential mutation automatically scales its steps with the dispersion of the population: when the population is spread out the steps are large (exploration) and, as it converges, the steps are reduced (exploitation), without needing to fix a constant mutation amplitude.

B. Genetic Algorithm

A real-coded GA is implemented with tournament selection (size 3), simulated binary crossover (SBX) [16] and polynomial mutation. SBX and polynomial mutation are operators suitable for continuous variables; their distribution indices, previously set by hand, are here determined by Optuna together with the rest of the hyperparameters. The hyperparameters tuned by Optuna are a population of 52 individuals, crossover probability 0.63, mutation probability 0.22, distribution indices $\eta_c = 16$ (SBX) and $\eta_m = 14$ (polynomial mutation) and elitism of 2 individuals. An early-stopping criterion by stagnation with a tolerance of 100 generations without improvement is applied; in the reported run the GA keeps improving until an advanced phase and exhausts the 300 generations without triggering that stop, which indicates that Optuna's tuning finds a configuration that keeps enough genetic diversity to keep exploring the search space throughout the assigned budget.

Table III
CONFIGURATION OF THE ALGORITHMS (HYPERPARAMETERS TUNED WITH OPTUNA)

Differential Evolution	**Genetic Algorithm**
strategy: best1bin	selection: tournament (3)
popsize: 28	population: 52
mutation: $(0.54, 0.85)$	crossover: SBX ($\eta_c = 16$)
recombination: 0.77	mutation: polynomial ($\eta_m = 14$)
maxiter: 300	generations: 300
tol: 10^{-7}	$p_{cx} = 0.63$, $p_{mut} = 0.22$
seed: 42	elite: 2, tolerance: 100
Tuning: Optuna (TPE), 25 trials per algorithm, seed 42	

V. RESULTS

A. Fitting error

Table IV gathers the metrics of both runs. DE obtains a global error of 2.17 mm, slightly lower than the 2.55 mm of the GA. The distribution by joint, illustrated in Figure 3, shows that DE fits the proximal phalanx better (PIP: 2.82 mm against 3.68 mm), while the GA, once tuned, reproduces the distal interphalangeal joint with greater fidelity (DIP: 1.43 mm against 1.53 mm of DE). Overall, the two solutions are very close in shape error, which confirms that both techniques are able to propagate the movement adequately up to the end of the mechanism.

Table IV
OPTIMIZATION RESULTS (SIMPLIFIED MODEL)

Metric	**DE**	**GA**
Global error (mm)	**2.1736**	2.5543
PIP error (mm)	**2.8156**	3.6813
DIP error (mm)	1.5317	**1.4273**
Fitness	**0.002335**	0.002720
Evaluations (nfev)	135 970	15 052
$\sum$ structural links (mm)	**347.2**	457.1

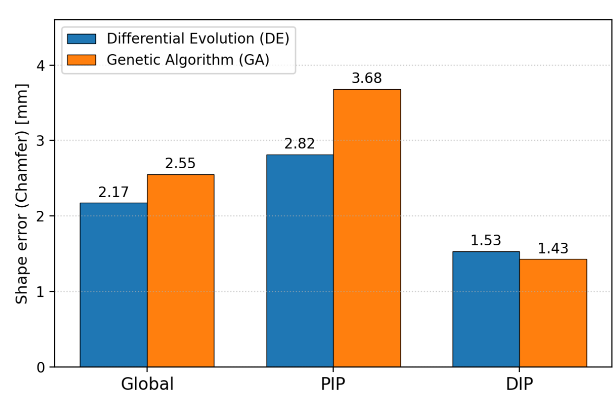

Figure 3. Shape error (Chamfer distance) global, at the PIP and at the DIP for DE and the GA. DE dominates in the global error and at the PIP, while the GA fits the DIP slightly better.

B. Dimensional analysis

Table V compares, link by link, the dimensions of the reference CAD design against those optimized by DE and the GA. The lengths obtained by DE are consistent with the bounds of Table I and give rise to a compact mechanism, where the sum of the eleven structural links (B_1, B_2, L_1–L_8 and c_2) is 347.2 mm for DE, 457.1 mm for the GA and approximately 380.0 mm for the base CAD design. That is, DE not only obtains a lower global error, but also delivers the most compact mechanism of the three, while the GA tends toward solutions larger than the reference design itself. Compactness is desirable for the manufacture, the weight and the portability of the device on the user's hand, so it constitutes a design criterion as relevant as the fitting error itself.

Table V
LINK DIMENSIONS (MM): BASE CAD DESIGN AGAINST THE OPTIMAL SOLUTIONS OF DE AND GA

Parameter	Base CAD	DE	GA
B_1	18.00	37.33	39.19
B_2	20.00	39.62	59.87
L_1	35.00	16.00	31.84
L_2	49.00	46.56	51.84
L_3	25.00	16.61	33.01
L_4	20.00	21.60	38.83
L_5	25.00	32.15	38.15
L_6	55.00	39.85	51.26
L_7	35.00	19.16	38.22
L_8	52.00	52.07	53.01
c_2	46.01	26.30	21.84
h_{sp}	17.00	14.91	15.63
d_{sp}	18.00	7.87	5.00
$\sum$ structural	380.0	**347.2**	457.1

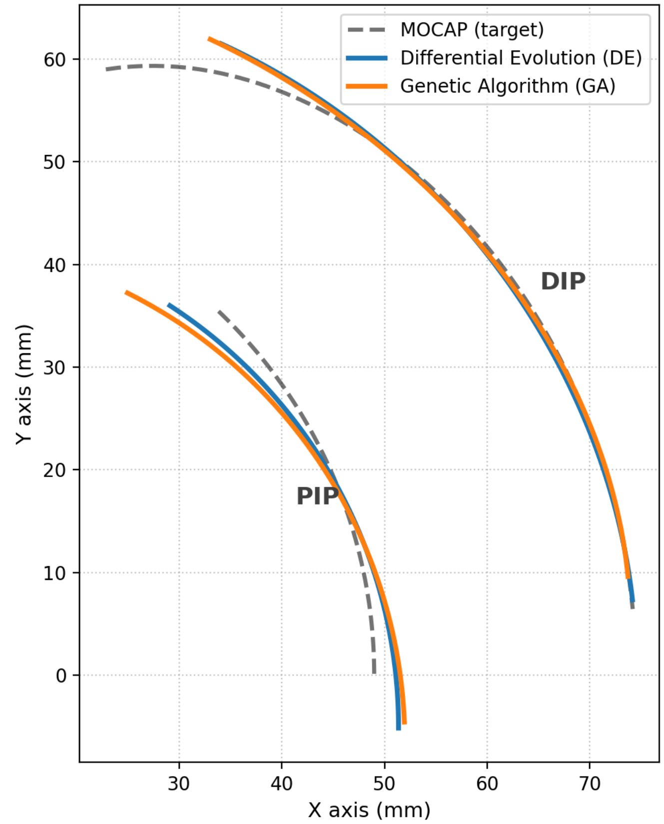

Figure 4. Trajectory biofidelity: simulated curves of the PIP and the DIP for DE and the GA, overlaid on the MOCAP points (target) of the fine pinch grasp.

C. Trajectory biofidelity

Figure 4 overlays the trajectories simulated by DE and by the GA on the 120 MOCAP points of the fine pinch grasp, both for the PIP and for the DIP. Both solutions follow the physiological shape closely: DE reproduces the PIP and the DIP with Chamfer errors of 2.82 mm and 1.53 mm, while the GA accompanies them with errors of 3.68 mm (PIP) and 1.43 mm (DIP). The DIP curve of the GA reproduces the natural flexion with greater fidelity, obtaining 0.10 mm less error than DE at that joint. Overall, both solutions exhibit a satisfactory biofidelity: DE offers better global tracking (2.17 mm versus 2.55 mm of the GA) and at the PIP, while the GA presents a specific advantage at the DIP.

D. Convergence

Figure 5 presents the convergence curves. Both algorithms descend steadily and use practically the whole number of available generations: DE records its last improvement near generation 257 and the GA near generation 268, so that neither stagnates early. DE reaches an objective function value of 0.002335, while the GA converges to 0.002720. It is worth noting that the GA arrives at that result with a considerably smaller number of objective function evaluations (Table IV), which reflects its different mechanism for sampling the search space.

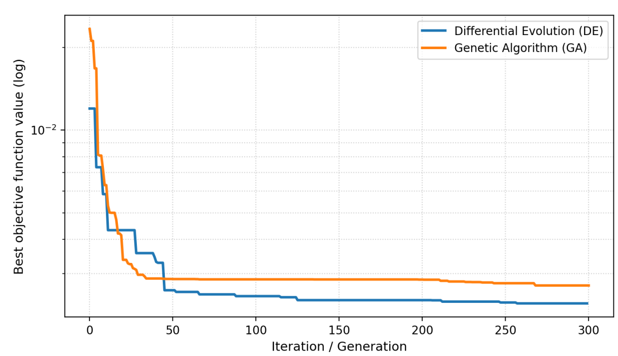

Figure 5. Convergence curves of DE and the GA. Both methods descend steadily during almost all the generations; DE converges to a fitness of 0.002335 and the GA to 0.002720.

VI. DISCUSSION

The design space of the dimensional synthesis is *continuous* and highly multimodal, with large infeasible regions (without assembly or with reversal of the medial phalanx) that generate plateaus in the objective function. The advantage of DE in this scenario comes from the way it builds each new solution: instead of applying a fixed step size, it adds to one individual the difference between two other individuals of the population. Thus, the magnitude of the change adapts by itself to how spread out the population is: when the individuals are distributed over the space, the changes are large and the algorithm explores; when the population concentrates near a good solution, the changes are reduced automatically and the algorithm refines the fit. This automatic step adjustment, without needing to fix it in advance, fits naturally with continuous variables such as the link lengths [6], [9] and explains, to a large extent, that DE reaches a global error of 2.17 mm and, above all, the most compact mechanism.

The real-coded GA relies on SBX crossover and polynomial mutation. Once its distribution indices (η_c, η_m) and its probabilities are tuned with Optuna, instead of being set by hand, the GA reaches a global error of 2.55 mm, very close to that of DE, and even surpasses it at the distal interphalangeal joint (DIP). This result shows that DE does not systematically dominate GAs in continuous problems [6], [7]: the observed gap is reduced notably when both algorithms start from a fair tuning. The trade-off of the GA in this case is that it tends toward larger solutions, which penalizes the compactness of the mechanism.

It is worth pointing out a methodological limitation regarding computational cost. DE is evaluated with a vectorized and parallel implementation, while the GA is run in pure Python; for this reason, the comparison of execution times is biased and is not reported as a merit criterion. Instead, the *quality of the solution* and the number of function evaluations (nfev) are emphasized as a fairer computational metric. DE reaches a slightly lower global error despite performing more evaluations, while the GA obtains a very close solution with a considerably smaller number of evaluations, which suggests that each method explores the search space with a different efficiency.

VII. Conclusion

The partial dimensional synthesis (up to the medial phalanx) of a finger exoskeleton mechanism is obtained using the optimization techniques Differential Evolution and real-coded Genetic Algorithm, under a single formulation, a single objective function and the same number of generations, and with the hyperparameters of both methods tuned symmetrically with Optuna (TPE sampler, 25 trials per algorithm). DE obtains a global error of 2.17 mm versus 2.55 mm of the GA and, notably, delivers the most compact mechanism (347.2 mm of sum of structural links, against 457.1 mm of the GA). Once tuned, the GA is competitive and even fits the distal interphalangeal joint (DIP) better. The analysis suggests that, for this class of continuous and multimodal problems, DE's differential mutation, which adjusts the size of its steps by itself, offers a slight advantage in global error and, above all, in compactness, while a well-tuned GA reaches a comparable fit quality. DE is recommended as a reference method when the compactness of the mechanism is a priority, without this ruling out the GA as a valid alternative for this class of dimensional synthesis.

As future work the following is considered: (i) extending the optimization to the full three-phalanx model, incorporating the distal phalanx and the third four-bar stage; (ii) carrying out multi-seed comparisons with statistical tests to estimate the robustness of each algorithm; (iii) implementing both algorithms under equal computational conditions (same language and same level of vectorization), so that their execution times can be compared without bias; and (iv) physically validating the optimal mechanism through prototyping and measurement of the real trajectories.

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